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Lewis structure]

Now let us calculate the formal charge on all atoms in both structures,

For Structure 1,

$$\text{Formal charge on carbon} = N_v - \left(N_l + \frac{N_b}{2}\right)$$

$$= 4 - \left(0 + \frac{8}{2}\right) = 0$$

$$\text{Formal charge on oxygen} = 6 - \left(4 + \frac{4}{2}\right)$$

$$= 0 \text{ (for both oxygens)}$$

For structure 2

Formal charge on carbon

$$= N_v - \left(N_l + \frac{N_b}{2}\right)$$

$$= 4 - \left(0 + \frac{8}{2}\right) = 0$$

Formal charge on singly bonded oxygen

$$= 6 - \left(6 + \frac{2}{2}\right) = -1$$

Formal charge on triply bonded oxygen

$$= 6 - \left(2 + \frac{6}{2}\right) = +1$$

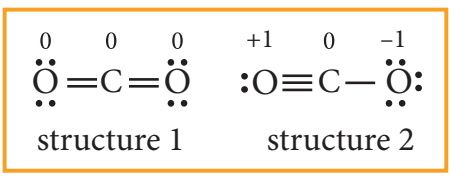


Fig 10. 7 (b) two possible structures for carbon dioxide (with formal charges)

After calculating the formal charges, the best representation of Lewis structure can be selected by using following guidelines.

1. A structure in which all formal charges are zero preferred over the one with charges.
2. A structure with small formal charges is preferred over the one with higher formal charges.
3. A structure in which negative formal charges are placed on the most electronegative atom is preferred.

In case of CO_2 structures, the structure one is preferred over the structure 2 as it has zero formal charges for all atoms.

10.2.4 Lewis structures for exceptions to octet rule

The octet rule is useful for writing Lewis structures for molecules with second period element as central atoms. In some molecules, the central atoms have less than eight electrons around them while some others have more than eight electrons. Exception to the octet rule can be categorized into following three types.

1. Molecules with electron deficient central atoms
2. Molecules containing odd electrons
3. Molecules with expanded valence shells

Molecules with electron deficient central atoms

Let us consider boron trifluoride, as an example. The central atom boron has three valence electron and each fluorine has seven valence electrons. The Lewis structure is

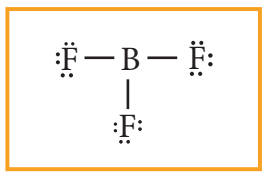


Fig 10. 8 (a) Lewis structure of BF_3

In the above structure, only six